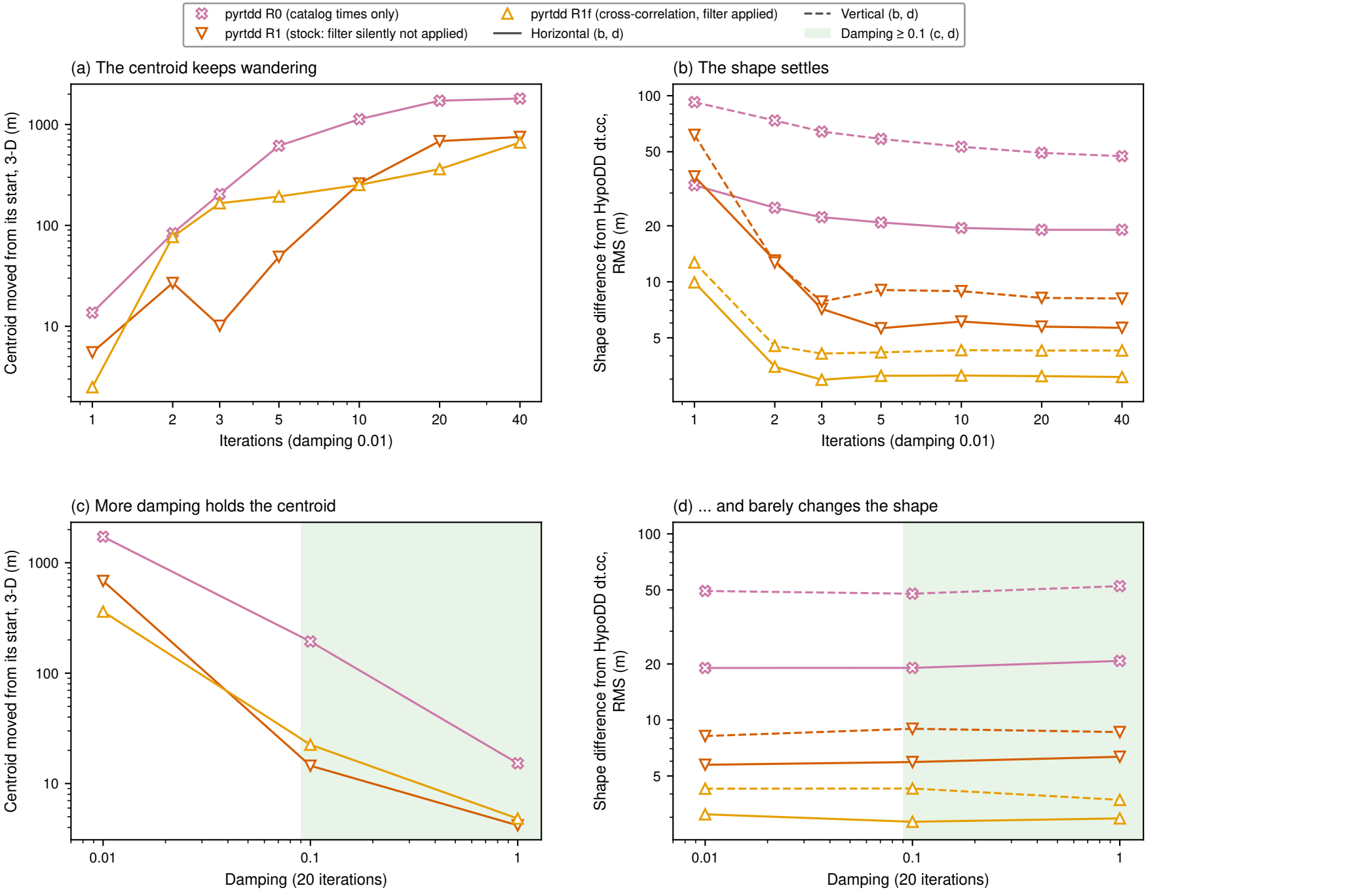
Pure double differences let the whole cluster slide (up to 1.7 km in 20 iterations at damping 0.01, R0) while its shape stays put; damping ≥ 0.1 keeps the cross-correlation runs within 22 m



Catalog-only R0 still moves 194 m at damping 0.1, 15 m at damping 1. HypoDD's SVD solver constrains the mean shift instead (its dt.cc centroid ended within 1.0 m of its start).

Shape difference from each run's own final shape (20 iterations, damping 0.01), RMS horizontal / vertical (m):

R0: 1 it. 24.4 / 60.8; 2 it. 14.0 / 37.6; 3 it. 9.7 / 26.5; 5 it. 6.6 / 18.5; 10 it. 2.7 / 7.6; damping 0.1: 2.7 / 7.0, damping 1: 6.4 / 15.8

R1: 1 it. 35.4 / 62.0; 2 it. 11.3 / 13.9; 3 it. 5.1 / 4.9; 5 it. 1.5 / 2.7; 10 it. 1.3 / 2.4; damping 0.1: 1.4 / 2.7, damping 1: 1.8 / 4.2

R1f: 1 it. 10.3 / 14.0; 2 it. 2.4 / 2.6; 3 it. 1.0 / 1.7; 5 it. 0.2 / 0.7; 10 it. 0.1 / 0.3; damping 0.1: 0.5 / 0.7, damping 1: 0.6 / 1.5